Risk Factors for Alcohol Consumption and Tobacco Use in Poor Households of Marikina City – A Pilot Study

REASEARCH PROPOSAL FOR STATISTICS 251-SURVEY OPERATIONS

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I. Background of the study

Smoking of tobacco and consumption of alcohol are considered two of the common behavioral causes for non-communicable diseases identified by the Department of Health (DOH, 2009). Tobacco smoking causes serious cardiovascular and respiratory diseases, including coronary heart disease and lung cancer for adults (WHO, 2016) which are the top causes of mortality in the Philippines (DOH, 2013). Harmful use of alcohol on the other hand causes more than 200 diseases and injury worldwide (WHO, 2015) including mental disorders, several types of cancer and cirrhosis (Villaverde et al, 2012).

The cost of the treatment for coronary heart disease and lung cancer are estimated to be at least Php 550,000 and Php 1,500,000 respectively (PhilHealth, 2014 and Rappler 2016) which are way beyond the annual income of an average Filipino family amounting to P267,000 as of 2015 (PSA, 2015). This makes the poor households more vulnerable to severe financial burdens once the above conditions are experienced.

This study aims to assess the prevalence of tobacco use and alcohol consumption in poor households and the risk factors that impact the behaviour towards tobacco and alcoholic beverages. At the end of the study, the researchers hope to identify the incidence of hazardous drinkers and smokers in the community, the potential relationship between tobacco use and alcohol consumption, and the factors that influence this trend which may lead to recommendations for mitigation of such risky behaviours.

II. Review of Related Literature

A. Tobacco Smoking

According to the 2009 Global Adult Tobacco Survey (GATS), 28.3% of the Filipinos are smoking tobacco, of which, 80% are daily smokers. The male comprise the majority of the smokers with 47.7% share from its population while the female group contributes only 9% from its population. Each consumes 11.3 and 7.0 cigarettes respectively on a daily basis. (GATS, 2009)

The study conducted by the Food and Nutrition Research Institute (FNRI) in 2013 shows a decline in the prevalence of smoking since 2008, however, the distribution of the incidence is

concentrated in the rural poor rather than the urban areas. The top three regions with the highest proportions of adult smokers are Zamboanga Peninsula, Luzon and CARAGA. (FNRI 2013)

Filipinos in general allocate 1.1% of their annual expenditure for tobacco while the lowest three income groups allocate 2% (PSA, 2015). This result paves the way to this study where the paper takes the opportunity to look closely at this trend with the focus on the lower income group and the cause of their behaviour that may be unique to them. The output aims to contribute to the mitigating steps in order to address the findings on tobacco use discussed above in the long run.

B. Alcohol consumption

Consumption of alcohol has become an acceptable practice imbibed in the history and culture of the Filipino society. Alcoholic beverages have been used as a tool for bonding, socializing, relaxing, and self-expression and self-disclosure by the Filipinos in general and drinking has been associated with happiness, success and sexual prowess (Labajo, 2010).

In the Asian region, Philippines was ranked as the third in terms of alcohol consumption following Thailand and Japan (Villaverde et al, 2012) consuming 3.5 liters of alcohol in 2001 per capita (WHO, 2004). In 2001, it was reported that 11.1% of the Filipinos are considered regular drinkers which is defined as those who drink for at least four days in a week; 38% are counted as occasional drinkers and 4.8% are heavy drinkers (Labajo, 2010).

The male population dominates the record of regular and heavy drinkers accounting to 13% and 7% of the population respectively compared to 6% and 1% for females. Alcohol is also shown to be prevalent to 39% of the Filipino youth ie. ages 15-29. (WHO 2004)

Among the variants of alcoholic beverages, spirits are shown to be the most consumed with an average consumption of around 3 liters per capita, followed by beer and wine. (WHO 2004)

Base from the Family Income and Expenditure Survey in 2015, the lower three income classes allot 0.8%-0.9% of their annual expenditure to purchasing alcoholic beverages. This is higher than the national average of 0.5% (PSA, 2015). Similar to the trends for tobacco, this paper also takes the opportunity to look closely at the trend for alcohol consumption with the focus on the

lower income group and their behaviour. The output also aims to contribute to the mitigating steps in order to address in the future the significant findings on alcohol consumption discussed above.

III. Statement of the Problem

The survey aims to serve as a pilot study to quantify the prevalence of tobacco smoking and alcohol consumption to poor households as well as assess the risk factors for these behaviors on both quantitative and qualitative approach. This will be done by collecting data from samples in all barangays of Marikina City which will include adults aged 18 and older coming from different households. The survey will be cross sectional and will be used to produce estimates at the city level.

Given the prevalence of tobacco smoking and alcohol consumption to the most vulnerable groups of the society, there is a need to narrow down the study to the poor households in order to have a better understanding of their unique behavioral patterns which will be used as a stepping stone to develop the most suitable strategies to mitigate the said behaviors.

IV. List of variables to be measured

The definitions of some of the variables are made consistent from the surveys conducted by WHO Global Status Report on Alcohol in 2004, Global Adult Tobacco Survey in 2009 and the Food and Nutrition Research Institute Report for Smoking Status of Filipino Children and Adults in 2013.

A. Gauge for Prevalence of Tobacco Smoking (Dependent Variables)

The term "tobacco" will be limited to smoked tobacco such as cigarettes including the electronic type, pipes and cigars. The term will exclude smokeless cigarettes which includes chewing of tobacco and betel quid.

Prevalence of tobacco smoking will be measured in terms of two variables – (1) the frequency and (2) the average number of sticks smoked in a day. Frequency will have three classifications: (1) Regular - if smoking is done daily; and (2) Occasional – if less frequent than daily; in this

case the actual number of days in a week/month will be asked (3) Never – if smoking is never done.

B. Gauge for Prevalence of Alcohol Consumption (Dependent Variables)

The term "alcohol" will refer to a psychoactive substance with dependence-producing properties. The study will limit the substance to the following types only: spirits, beer and wine.

Prevalence of alcohol consumption will be measured in terms of two variables – (1) the frequency and (2) the average number of 250 ml bottles consumed in a day. Frequency will have three classifications: (1) Regular - if drinking is done for at least four days a week; (2) Occasional – if less frequent than four days a week; in this case the actual number of days in a week/month will be asked and (3) Never – if drinking is never done.

C. Potential Risk Factors (Independent Variables)

These variables represent the factors that will be tested in terms of their impact to the behavior towards alcohol and tobacco.

i. Educational Background –the level of education completed by the respondent (dummy variable).

ii. Influence from Family Members– "family members" is defined as the nuclear and extended family members of the respondent which includes the mother, the father, the siblings, the grandparents, the aunts, the uncles, and other relatives, who all live nearby or in one household. This variable will be measured by the number of family members who are smokers/drinkers (count data) and how many of them influenced the respondent's decision (directly or indirectly) to/not to smoke or drink (count data).

iii. Influence from Peers – "peers" is defined as the individuals that are not family members whom the respondent considers as companions and whom he spends time with for at least an hour a week. This may include the respondent's friends or officemates. This variable will be measured by the number of peers who are

smokers/drinkers (count data) and how many of them influenced the respondent's decision (directly or indirectly) to/not to smoke or drink (count data).

iv. Availability - ease of access as determined by:

a. Price Tolerance – the maximum amount the person is willing to pay in exchange of a unit of tobacco or an alcoholic beverage. The variable will be measured by the maximum amount the respondent is willing to pay for a unit of tobacco or alcoholic beverage. This will be grouped per price bracket (dummy variable).

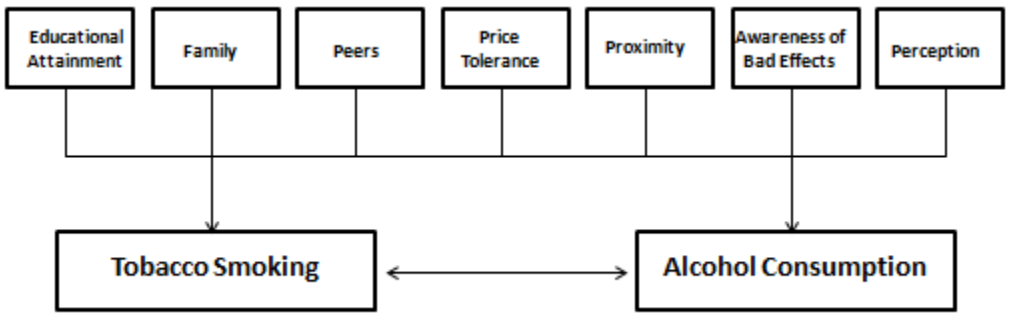
b. Proximity – the distance of the nearest distributor where the respondent purchases tobacco and alcoholic beverage in meters. The respondent will be free to state his inputs (continuous variable).

v. Awareness of the Negative Effects of Smoking and Alcohol to Health – whether the respondent is aware of the negative effects of smoking and alcohol to health as discussed in the background. The answer will either be a yes or a no (dummy variable).

vi. Perceptions towards Smoking/Drinking –how the respondent perceives the effects of smoking and drinking to his psychological, emotional and physical well being. This will be assessed on a qualitative level where the respondent will be free to state his inputs.

V. Schematic Diagram

The relationship of the variables enumerated is illustrated as follows:



VI. Limitations of the study

Due to the limitation of time and resources, the study will only target to produce estimates for Marikina City. Samples from the non-residents and nationwide sampling will not be considered. The study will only focus on adults and will not cover the youth and the juvenile cases. The samples will not be monitored over time.

VII. How are you going to collect the data?

A cross-sectional household survey will be conducted in collecting the data. The aim is to acquire estimates of basic statistics for all the dependent variables defined at the city level in order to arrive at an accurate summary of the city profile. Another aim is to get complete and accurate data for the independent variables in order to establish reliable relationships between the behaviour and risk factors.

VIII. Statistical Procedure for the study

A. Hypotheses to be tested

There are several hypotheses to be tested in the study as follows:

i. More than 23% of the poor households are considered regular smokers
ii. More than 11% of the poor households are considered regular drinkers
iii. Tobacco smoking and alcohol consumption are positively correlated
iv. Educational Attainment is a significant risk factor for tobacco smoking in terms of frequency and average no. of sticks smoked per day
v. Educational Attainment is a significant risk factor for consuming alcohol in terms of frequency and average no. of bottles consumed per day
vi. Influence from Family is a significant risk factor for tobacco smoking in terms of frequency and average no. of sticks smoked per day
vii. Influence from Family is a significant risk factor for consuming alcohol in terms of frequency and average no. of bottles consumed per day
viii. Influence from Peers is a significant risk factor for tobacco smoking in terms of frequency and average no. of sticks smoked per day

ix. Influence from Peers is a significant risk factor for consuming alcohol in terms of frequency and average no. of bottles consumed per day
x. Price Tolerance is a significant risk factor for tobacco smoking in terms of frequency and average no. of sticks smoked per day
xi. Price Tolerance is a significant risk factor for consuming alcohol in terms of frequency and average no. of bottles consumed per day
xii. Proximity is a significant risk factor for tobacco smoking in terms of frequency and average no. of sticks smoked per day
xiii. Proximity is a significant risk factor for consuming alcohol in terms of frequency and average no. of bottles consumed per day
xiv. Perception towards Smoking/Drinking is a risk factor for tobacco smoking in terms of frequency and average no. of sticks smoked per day
xv. Perception towards Smoking/Drinking is a risk factor for consuming alcohol in terms of frequency and average no. of bottles consumed per day

B. Target Population

The target population of the study includes all men and women, ages 18 and above who consider Marikina City as their permanent place of residence and falls within the three lowest income class as defined in the Family Income and Expenditure Survey which is below Php 99,999 per year.

C. Sampling Design and Sample Size

A two-staged sample design will be considered where the Primary Sampling Units will be the barangays. The PSUs will be selected using probability proportional to the estimated number of poor households in a barangay. This number will be estimated as 5% of the barangay population. The 5% estimate was derived from the actual proportion of poor families to the population in Metro Manila. From the selected barangays, the second-stage sampling will be done by selecting housing units through systematic sampling. Target proportion for male and female is 50% ie. one male respondent should be interviewed for every female.

The sampling size was computed using the formula for proportions. For this case, both the scenarios for alcohol consumption and tobacco smoking were considered. The most conservative estimate was chosen in the end.

i. Tobacco Smoking

Assumptions:
P: Incidence of Tobacco Smoking = 0.28
Q: Incidence of Non-smoking = 0.72
N: Number of Households in Marikina that will fall under the lowest three income brackets
= Number of Households in Marikina (Marikina City Website, 2013) * Proportion of households in Metro Manila that will fall under the lower three income bracket (PSA, 2015)
= 112,727[1] * 0.05[2]
= 5,408
$Z_{a/2}$: Risk Assumed = 1.96
d: Error Assumed = 0.05

$$n = \frac{(Z_{a/2})^2 (NPQ)/(N-1)}{d^2 + (Z_{a/2})^2 (PQ)/(N-1)}$$

n = 293

ii. Alcohol Consumption

Assumptions:
P: Incidence of Tobacco Smoking = 0.11
Q: Incidence of Non-smoking = 0.89
N: Number of Households in Marikina that will fall under the lowest three income brackets
= Number of Households in Marikina (Marikina City Website, 2013) * Proportion of households in Metro Manila that will fall under the lower three income bracket (PSA, 2015)
= 112,727[1] * 0.05[2]
= 5,408
$Z_{a/2}$: Risk Assumed = 1.96
d: Error Assumed = 0.05

$$n = \frac{(Z_{a/2})^2 (NPQ)/(N-1)}{d^2 + (Z_{a/2})^2 (PQ)/(N-1)}$$

n = 146

The results above show that the sample should be at least 293 households in order to yield significance of at least 95% and assumed error of at most 5% for both tobacco smoking and alcohol consumption.

D. Statistical Analysis

i. Descriptive Statistics and Associated Dummy Variables

The table below summarizes the characteristics of the population based on the frequency of tobacco smoking, alcohol consumption and their combination.

	Tobacco Smoking			Alcohol Consumption			Combined			
	A. Regular	B. Non-Reg	C. Never	D. Regular	E. Non-Reg	F. Never	Both A D	Both A E	Both B D	Both B E
Total										
Proportion										
Total Male										
Prop Male										
Total Female										
Prop Female										
Average Age										

ii. Tests of Hypothesis and Associated Dummy Variable

The table below summarizes the relationships of the dependent and independent variables by showing the mean values of each of the risk factors per frequency measures and average sticks/bottles. Relationship between tobacco smoking and alcohol consumption will also be shown.

Tobacco Smoking

	Education	Family	Peers	Price Tolerance	Proximity	Awareness of Effects	Alcohol Freq	Alcohol Bottles
Regular								
Non-Reg								
Never								

	Education	Family	Peers	Price Tolerance	Proximity	Awareness of Effects	Alcohol Freq	Alcohol Bottles
Average Sticks								

Alcohol Consumption

	Education	Family	Peers	Price Tolerance	Proximity	Awareness of Effects	Tobacco Freq	Tobacco Sticks
Regular								
Non-Reg								
Never								

	Education	Family	Peers	Price Tolerance	Proximity	Awareness of Effects	Tobacco Freq	Tobacco Sticks
Average Bottles								

Other factors that may affect the behaviour towards tobacco smoking and alcohol consumption including perception will be asked and analyzed qualitatively.

iii. Statistical modelling

Regression analysis will be used in quantifying the actual impact of the risk factors to the dependent variables as well as the relationship between tobacco smoking and alcohol consumption.

Tobacco Smoking

Y1	X1	X2	X3	X4	X5	X6	X7	
Tobacco Frequency	**Education**	**Family**	**Peers**	**Price Tolerance**	**Proximity**	**Awareness of Effects**	**Alcohol Freq**	**Alcohol Bottles**

Y2	X1	X2	X3	X4	X5	X6	X7	
Average Sticks	**Education**	**Family**	**Peers**	**Price Tolerance**	**Proximity**	**Awareness of Effects**	**Alcohol Freq**	**Alcohol Bottles**

Alcohol Consumption

Y1	X1	X2	X3	X4	X5	X6	X7	
Alcohol Frequency	**Education**	**Family**	**Peers**	**Price Tolerance**	**Proximity**	**Awareness of Effects**	**Tobacco Freq**	**Tobacco Sticks**

Y2	X1	X2	X3	X4	X5	X6	X7	
Alcohol Bottles	**Education**	**Family**	**Peers**	**Price Tolerance**	**Proximity**	**Awareness of Effects**	**Tobacco Freq**	**Tobacco Sticks**

IX. Proposed Budget

A. Overhead Cost

i. Office Rental

Survey operation is planned to run from February to May 2017. Brgy. Sta. Elena is a preferred location since it is located at the center of the city. Considering an average renting cost of Php 4,000 in Brgy. Sta. Elena for a 25 sqm studio type apartment, the total budget for the office rental will be Php 16,000.

ii. Office and Field Equipments

Office and field equipments considered are the minimum resources for the daily operations including trainings. Detailed breakdown is shown below.

	No. of Units	Unit Cost	Total Cost
Utilities – Water	4 mos	500	2,000
Utilities – Electricity	4 mos	500	2,000
Internet Device	1	2,000	2,000
Monthly Internet	4 mos	1,000	4,000
Office Tables	2	1,000	2,000
Monobloc Chairs	10	300	3,000
Electric Fan	2	500	1,000
Laptop	1	20,000	20,000
Printer and Photocopier	1	8,000	8,000
Toners	3 mos	2,000	6,000
Projector	1	5,000	5,000
Mobile Phone	1	5,000	5,000
Mobile Load	1	500	500
Miscellaneous Supplies	4 mos	2,000	8,000
Grand Total			68,500

iii. Salary of Staff and Interviewers

Data gathering is expected to run for two weeks. Assuming each questionnaire will take a maximum of one hour to fill out, and at least four qualified respondents will be interviewed in a day, it will take at least five respondents to complete 293 samples in two weeks.

The salary will be based on the number of qualified respondents interviewed. Training allowance will also be provided per interviewer.

A supervisor will be assigned to monitor the operational and logistical needs during the fieldwork as well as to make sure that the qualifications and parameters set are followed. He will be doing the overall supervision from the conceptualization phase, encode and analyze the survey results and translate them in a form of a report.

	No. of Units	Unit Cost	Total Cost
Field Interviewers			
Interview Salary (per qualified respondent)	293	500	146,500
Training Allowance (per interviewer)	5	300	1,500
Supervisor			
Salary (per month)	3	13,000	39,000
Grand Total			**187,000**

B. Transportation and Communication

Respondents for a maximum of two barangays will be considered in a day to achieve efficiency. Given this and considering the current public transportation fare of Php 7 pesos via jeepney, the proposed budget for daily transportation is Php 50, allowing for buffer. This will yield a total of Php 700 cost per interviewer and Php 3,500 in total for all. Interviewers are expected to submit to the office their reports daily.

The interview will be using pen and paper, hence no budget for communication will be provided. Minimal need to communicate to the supervisor is factored in the salary. In most cases, the supervisor is responsible in contacting the field interviewers using the office resources.

C. Training and Field Work

i. Training

Training will be done in house and may run for 2-3 hours. A budget for snacks amounting to Php 50 per person or Php 250 for all will be allotted. Training allowance is already incorporated in the table presented for salary.

iii. Interviewer kits

Printing of the questionnaires and pens to be used are already incorporated in the office supplies and equipments' budget.

Token for the interviewers amounting to Php 200 per person will be provided which will cost Php 58,600 for all 293 respondents.

D. Editing and Data Processing

Editing and data processing will be done in house by the supervisor using the office resources. Stata v13 will be used and will cost at least Php 10,000 assuming no software is available for use yet.

E. Report Writing

i. Reproduction

The cost for reproduction will be included in the budget for miscellaneous supplies. The original copy will be kept by the primary sponsor. The report will be reproduced mainly electronically.

ii. Binding

The cost for binding will be included in the budget for miscellaneous supplies.

iii. Mailing

The cost for mailing will be included in the budget for miscellaneous supplies. The report will be transported more often electronically.

X. Gantt Chart

2017	Feb wk 2	Feb wk 3	Feb wk 4	Mar wk 1	Mar wk 2	Mar wk 3	Mar wk 4	Apr wk 1	Apr wk 2	Apr wk 3	Apr wk 4	May wk 1	May wk 2	May wk 3	May wk 4	Jun wk 1	Jun wk 2	Jun wk 3	Jun wk 4
Finalizing Research Proposal	■	■	■																
Developing the Questionnaire				■	■														
Testing the Questionnaire						■													
Securing Logistics				■	■	■	■												
Training								■	■										
Data Gathering										■	■								
Data Processing and Aggregation												■							
Report Writing													■						
Submission														■					

XI. Questionnaire

High level draft and outline of the questionnaire is shown below:

1. Respondent Number:____
2. How old are you:____ (if at least 18 y/o proceed to 3, otherwise do not proceed)
3. What is your highest educational attainment to date?_____
4. Are you permanently residing in <Insert name of baranggay>?___ (if yes proceed to 4, , otherwise do not proceed)
5. How many are you in your household? (No. of individuals he stays with in their primary residence)
6. How much is your household gross income in a year?_____ (if below P99,999 then proceed to A.1, otherwise do not proceed)

A. Tobacco Use

A.1 Have you tried smoking tobacco in the form of cigarettes, cigars, pipes or e-cigarettes?___ (if yes, proceed to A.2, otherwise proceed to A.5)

A.2 If yes, which among the ff have you tried? Tick whichever applies:
☐ Cigarette
☐ Cigar
☐ Pipe
☐ e-Cigarette
Others:_____

A.3 How often do you smoke? Tick only one:
☐ Regular (Daily)
☐ Occasional (less frequent than daily). How many days in a week/month?___

A.4 How many sticks (for cigarettes and cigars)/dosage (for pipes and e-Cigarettes)/units (for others) do you consume in a day of use?
Cigarette:___
Cigar:___
Pipe:___
e-Cigarette:___
Others:_____

A.5 How many of your family members (Nuclear and extended within the same household) have smoked?_____
A.5.1 Who are they in specific (mention only relationship)?_____

A.6.1 Does any of your family members influence your decision to smoke/not smoke?___
A.6.2 Who are they in specific (mention only relationship)?_____
A.6.3 In what way?_____

A.7.1 How many of your peers (individuals whom the respondent spends with at least an hour a week) have smoked?_____
A.7.2 Who are they in specific (mention only relationship)?_____

A.8.1 Does any of your peers influence your decision to smoke/not smoke?___
A.8.2 Who are they in specific (mention only relationship)?_____
A.8.3 In what way?_____

A.9.1 (For smokers only) How much are you currently buying a stick (for cigarettes and cigars)/dosage (for pipes and e-Cigarettes)/unit (for others)?_____
A.9.2 How much is the maximum price you are willing to buy a stick/dosage/unit?_____

A.10.1 Where is the nearest place/store/establishment you buy your stick/dosage/unit? _____
A.10.2 How far is it from your home (in meters)?_____

A.11 What do you think are the effects of smoking to you physically, emotionally, and emotionally?_____
_____

A.12 Do you know that smoking causes diseases such as heart failure, cancer and stroke?___
A. 13 What are your other reasons for smoking/not smoking?_____
_____

B. Alcohol Consumption

B.1 Have you tried drinking alcoholic beverages in the form of beer, spirits or wine?___ (if yes, proceed to B.2, otherwise proceed to B.5)

B.2 If yes, which among the ff have you tried? Tick whichever applies:
☐ Beer
☐ Spirits
☐ Wine
Others:_____

B.3 How often do you drink? Tick only one:
☐ Regular (Daily)
☐ Occasional (less frequent than daily). How many days in a week/month?___

B.4 How many bottle (250 ml) do you consume in a day of drinking?
Beer:___
Spirits:___
Wine:___
Others:_____

B.5 How many of your family members (Nuclear and extended within the same household) have tried drinking?_____
B.5.1 Who are they in specific (mention only relationship)?_____

B.6.1 Does any of your family members influence your decision to drink/not drink?___
B.6.2 Who are they in specific (mention only relationship)?_____
B.6.3 In what way?_____

B.7.1 How many of your peers (individuals whom the respondent spends with at least an hour a week) have tried drinking?_____
B.7.2 Who are they in specific (mention only relationship)?_____

B.8.1 Does any of your peers influence your decision to drink/not drink?___
B.8.2 Who are they in specific (mention only relationship)?_____
B.8.3 In what way?_____

B.9.1 (For drinkers only) How much are you currently buying a bottle (250 ml)?_____
B.9.2 How much is the maximum price you are willing to buy a bottle (250 ml)?_____

B.10.1 Where is the nearest place/store/establishment you buy your bottle?
_____
B.10.2 How far is it from your home (in meters)?_____

B.11 What do you think are the effects of drinking to you physically, emotionally, and emotionally?_____
_____

B.12 Do you know that drinking causes diseases such as heart failure, cancer and stroke?___
B. 13 What are your other reasons for drinking/not drinking?_____
_____

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